

Supplementary Materials

Supplementary Appendix A. The Proof of Proposition 1. This appendix provides a detailed proof of *Proposition 1* in Section III-B1 of the paper.

First, it is proved that the swarm avoids stagnation when $0 < \alpha \leq 1$. Since the swarm is confined by the tube boundaries in the d_n and d_b directions, there is no swarm inflow or outflow across the tube boundaries. Under the continuity constraint (31) of the manuscript, it follows that

$$\rho_{g,l}(l)v_{d,l}(l)S(l) = Q_0, \quad (1)$$

where $Q_0 > 0$ is a constant. Substituting the desired density in Eq. (17) of the manuscript,

$$\rho_{g,l}(l) = \rho_0 \left(\frac{S(l)}{S_0} \right)^{-\alpha}, \quad (2)$$

into the above Eq. (1) yields

$$v_{d,l}(l) = \frac{Q_0}{\rho_{g,l}(l)S(l)} = \frac{Q_0}{\rho_0 \left(\frac{S(l)}{S_0} \right)^{-\alpha} S(l)} = \frac{Q_0}{\rho_0 S_0^\alpha} S(l)^{\alpha-1}. \quad (3)$$

Let

$$C_1 = \frac{Q_0}{\rho_0 S_0^\alpha} > 0. \quad (4)$$

Then,

$$v_{d,l}(l) = C_1 S(l)^{\alpha-1}. \quad (5)$$

Since the cross-sectional area satisfies

$$0 < S_{\min} \leq S(l) \leq S_{\max} < \infty, \quad (6)$$

and $0 < \alpha \leq 1$, it follows that $\alpha - 1 \leq 0$. Therefore,

$$v_{d,l}(l) = C_1 S(l)^{\alpha-1} \geq C_1 S_{\max}^{\alpha-1} > 0. \quad (7)$$

Hence, the planned forward velocity is bounded away from zero, and the swarm avoids stagnation inside the virtual tube.

Next, it is proved that the desired density does not exceed the maximum density when $\alpha \leq \frac{\ln(\rho_{\max}/\rho_0)}{\ln(S_0/S_{\min})}$. Since $S_{\min} \leq S_0$, given the range of α , it implies that

$$-\alpha \ln \left(\frac{S_{\min}}{S_0} \right) \leq \ln \left(\frac{\rho_{\max}}{\rho_0} \right). \quad (8)$$

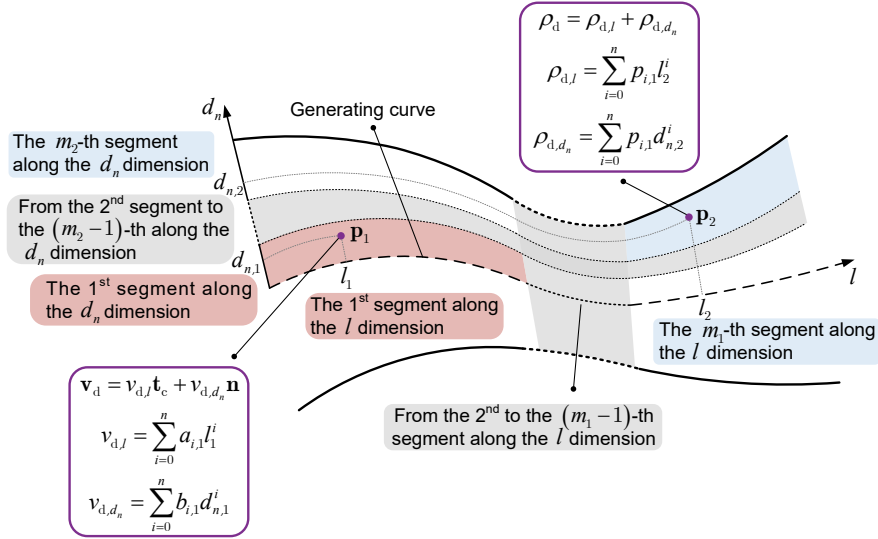
According to Eq. (17) of the manuscript, it is easily derived that

$$\rho_{g,l} \leq \rho_0 \left(\frac{S_{\min}}{S_0} \right)^{-\alpha} \leq \rho_{\max}. \quad (9)$$

This completes the proof.

Supplementary Appendix B. Mathematical Representation of VFF Planning. This appendix provides a detailed mathematical description of the VFF planning method, with the key symbols defined in Section III-A and Section III-B of the manuscript.

According to Eqs. (14) and (15) of the manuscript, $v_{d,l}$, v_{d,d_n} , v_{d,d_b} , $\rho_{d,l}$, ρ_{d,d_n} , and ρ_{d,d_b} are represented by n -th order



Supplementary Fig.1. Schematic illustration of piecewise polynomial planning for a VFF in a 2D space. The polynomial forms of the desired velocity $\mathbf{v}_d(\mathbf{p}_1)$ at position $\mathbf{p}_1$ and the desired density $\rho_d(\mathbf{p}_2)$ at position $\mathbf{p}_2$ are shown. For the sake of brevity, $\mathbf{p}_1, \mathbf{p}_2$ have been omitted.

piecewise polynomials as

$$\begin{aligned}
 v_{d,l}(l) &= f_{m_1}(l, a), \\
 v_{d,d_n}(d_n) &= f_{m_2}(d_n, b), \\
 v_{d,d_b}(d_b) &= f_{m_3}(d_b, c), \\
 \rho_{d,l}(l) &= f_{m_1}(l, p), \\
 \rho_{d,d_n}(d_n) &= f_{m_2}(d_n, w), \\
 \rho_{d,d_b}(d_b) &= f_{m_3}(d_b, z),
 \end{aligned} \tag{10}$$

$$f_{m_j}(q, k) = \begin{cases} \sum_{i=0}^n k_{i,1} q^i & q_0 \leq q < q_1 \\ \sum_{i=0}^n k_{i,2} q^i & q_1 \leq q < q_2 \\ \vdots & \vdots \\ \sum_{i=0}^n k_{i,m_j} q^i & q_{m_j-1} \leq q \leq q_{m_j} \end{cases},$$

where $j = 1, 2, 3$, the variable $k_{i,g}$ is the coefficient of the i -th order term in the g -th polynomial, where $g = 1, 2, \dots, m_j$, and m_j represents the number of segments. Supplementary Fig.1 illustrates the piecewise polynomial representations of $\mathbf{v}_d(\mathbf{p}_1)$ and $\rho_d(\mathbf{p}_2)$ in a 2D scenario.

Next, define the coefficient vector of the polynomial as follows,

$$\mathbf{a}_j = [a_{0,j} \quad a_{1,j} \quad \dots \quad a_{n,j}]^T, \tag{11}$$

where $j = 1, 2, \dots, m_1$ represents the j -th segment. Similarly, define coefficient vectors $\mathbf{b}_j, \mathbf{c}_j, \mathbf{p}_j, \mathbf{w}_j, \mathbf{z}_j$. Then, define

$$\mathbf{x}_1 = [\mathbf{a}_1^T \quad \mathbf{a}_2^T \quad \dots \quad \mathbf{a}_{m_1}^T]^T. \tag{12}$$

Similarly, define $\mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_6$ according to $\mathbf{b}_j, \mathbf{c}_j, \mathbf{p}_j, \mathbf{w}_j, \mathbf{z}_j$ respectively. Then, the optimization vector is $\mathbf{x} = [\mathbf{x}_1^T \quad \mathbf{x}_2^T \quad \mathbf{x}_3^T \quad \mathbf{x}_4^T \quad \mathbf{x}_5^T \quad \mathbf{x}_6^T]^T$.

Some vectors are further defined as follows,

$$\begin{aligned}
 \mathbf{l}_1 &= [1 \quad l \quad l^2 \quad l^3 \quad \dots \quad l^n]^T \\
 \mathbf{l}_2 &= [0 \quad 1 \quad 2l \quad 3l^2 \quad \dots \quad nl^{n-1}]^T
 \end{aligned} \tag{13}$$

Similarly, define vectors $\mathbf{d}_{n,1}, \mathbf{d}_{n,2}, \mathbf{d}_{b,1}, \mathbf{d}_{b,2}$.

Next, to enhance clarity, several matrices are defined as follows,

$$\mathbf{L}_1 = \begin{bmatrix} \mathbf{1}_1^T & \mathbf{0}_{1 \times (n+1)} & \cdots & \mathbf{0}_{1 \times (n+1)} \\ \mathbf{0}_{1 \times (n+1)} & \mathbf{1}_1^T & \cdots & \mathbf{0}_{1 \times (n+1)} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{0}_{1 \times (n+1)} & \mathbf{0}_{1 \times (n+1)} & \cdots & \mathbf{1}_1^T \end{bmatrix}_{m_1 \times m_1(n+1)},$$

The subscript $m_1 \times m_1(n+1)$ in the lower right corner indicates that the matrix consists of m_1 rows and $m_1(n+1)$ columns. Similarly, matrices $\mathbf{L}_2, \mathbf{D}_{n,1}, \mathbf{D}_{n,2}, \mathbf{D}_{b,1}, \mathbf{D}_{b,2}$ are defined based on $\mathbf{l}_2, \mathbf{d}_{n,1}, \mathbf{d}_{n,2}, \mathbf{d}_{b,1}, \mathbf{d}_{b,2}$ respectively. Then, define

$$\tilde{\mathbf{L}}_1 = \begin{bmatrix} \mathbf{1}_1^T & -\mathbf{1}_1^T & \mathbf{0}_{1 \times (n+1)} & \cdots & \mathbf{0}_{1 \times (n+1)} \\ \mathbf{0}_{1 \times (n+1)} & \mathbf{1}_1^T & -\mathbf{1}_1^T & \cdots & \mathbf{0}_{1 \times (n+1)} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ \mathbf{0}_{1 \times (n+1)} & \mathbf{0}_{1 \times (n+1)} & \mathbf{0}_{1 \times (n+1)} & \cdots & -\mathbf{1}_1^T \end{bmatrix},$$

where $\tilde{\mathbf{L}}_1$ and $\mathbf{L}_1$ are of the same size. Similarly, matrices $\tilde{\mathbf{L}}_2, \tilde{\mathbf{D}}_{n,1}, \tilde{\mathbf{D}}_{n,2}, \tilde{\mathbf{D}}_{b,1}, \tilde{\mathbf{D}}_{b,2}$ are defined based on $\mathbf{l}_2, \mathbf{d}_{n,1}, \mathbf{d}_{n,2}, \mathbf{d}_{b,1}, \mathbf{d}_{b,2}$ respectively. Subsequently, define $\mathbf{A}_1 = \text{diag}(\mathbf{L}_1, \mathbf{D}_{n,1}, \mathbf{D}_{b,1}, \mathbf{0}, \mathbf{0}, \mathbf{0})_{M_1 \times M_1(n+1)}$, where $\text{diag}(\ast)$ represents the construction of a diagonal matrix, $M_1 = 2(m_1 + m_2 + m_3)$, and $\mathbf{0} = \mathbf{0}_{m_1 \times m_1(n+1)}, \mathbf{0} = \mathbf{0}_{m_2 \times m_2(n+1)}, \mathbf{0} = \mathbf{0}_{m_3 \times m_3(n+1)}$ in $\mathbf{A}_1$ respectively. Define

$$\hat{\mathbf{L}}_1 = \mathbf{L}_1 \otimes \mathbf{1}_{m_2} \otimes \mathbf{1}_{m_3},$$

$$\hat{\mathbf{D}}_{n,1} = \mathbf{1}_{m_1} \otimes \mathbf{D}_{n,1} \otimes \mathbf{1}_{m_3},$$

$$\hat{\mathbf{D}}_{b,1} = \mathbf{1}_{m_2} \otimes \mathbf{1}_{m_3} \otimes \mathbf{D}_{b,1},$$

where $\mathbf{1}_{m_1}$ denotes the m_1 -dimensional column vector of ones. Similarly, matrices $\hat{\mathbf{L}}_2, \hat{\mathbf{D}}_{n,2}, \hat{\mathbf{D}}_{b,2}$ are defined. Define $\mathbf{B}_1 = [\mathbf{0} \ \mathbf{0} \ \mathbf{0} \ \hat{\mathbf{L}}_1 \ \hat{\mathbf{D}}_{n,1} \ \hat{\mathbf{D}}_{b,1}]$, $\mathbf{B}_2 = [\hat{\mathbf{L}}_2 \ \hat{\mathbf{D}}_{n,2} \ \hat{\mathbf{D}}_{b,2} \ \mathbf{0} \ \mathbf{0} \ \mathbf{0}]$, $\mathbf{B}_3 = \text{diag}(\hat{\mathbf{L}}_1, \hat{\mathbf{D}}_{n,1}, \hat{\mathbf{D}}_{b,1}, \mathbf{0}, \mathbf{0}, \mathbf{0})$, $\mathbf{B}_4 = \text{diag}(\hat{\mathbf{L}}_2, \hat{\mathbf{D}}_{n,2}, \hat{\mathbf{D}}_{b,2}, \mathbf{0}, \mathbf{0}, \mathbf{0})$, $\mathbf{B}_5 = \text{diag}(\mathbf{0}, \mathbf{0}, \mathbf{0}, \hat{\mathbf{L}}_1, \hat{\mathbf{D}}_{n,1}, \hat{\mathbf{D}}_{b,1})$, $\mathbf{B}_6 = \text{diag}(\mathbf{0}, \mathbf{0}, \mathbf{0}, \hat{\mathbf{L}}_2, \hat{\mathbf{D}}_{n,2}, \hat{\mathbf{D}}_{b,2})$, $\mathbf{B}_7 = \text{diag}(\hat{\mathbf{L}}_1, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0})$, $\mathbf{C}_1 = \text{diag}(\tilde{\mathbf{L}}_1, \tilde{\mathbf{D}}_{n,1}, \tilde{\mathbf{D}}_{b,1}, \tilde{\mathbf{L}}_1, \tilde{\mathbf{D}}_{n,1}, \tilde{\mathbf{D}}_{b,1})$, $\mathbf{C}_2 = \text{diag}(\tilde{\mathbf{L}}_2, \tilde{\mathbf{D}}_{n,2}, \tilde{\mathbf{D}}_{b,2}, \tilde{\mathbf{L}}_2, \tilde{\mathbf{D}}_{n,2}, \tilde{\mathbf{D}}_{b,2})$. The aforementioned matrices differ in their row dimensions but share a common column dimension of $M_1(n+1)$. In particular, the row dimension is M_2 for $\mathbf{B}_1$ and $\mathbf{B}_2$, $6M_2$ for $\mathbf{B}_3$ through $\mathbf{B}_6$, and M_1 for $\mathbf{C}_1$ and $\mathbf{C}_2$, where $M_2 = m_1 m_2 m_3$, and $\mathbf{0} = \mathbf{0}_{M_2 \times m_1(n+1)}, \mathbf{0} = \mathbf{0}_{M_2 \times m_2(n+1)}, \mathbf{0} = \mathbf{0}_{M_2 \times m_3(n+1)}$ in $\mathbf{B}_j$, where $j = 1, 2, \dots, 6$. Additionally, let $\mathbf{A}_{1,j}$ denote the j -th row of $\mathbf{A}_1$, where $j = 1, 2, \dots, M_1$.

Therefore, based on the first inequality in Eq. (26) of the manuscript, the linear inequality constraint of problem (32) of the manuscript can be derived as

$$\begin{bmatrix} \mathbf{B}_1 \\ -\mathbf{B}_1 \end{bmatrix} \mathbf{x} \leq \begin{bmatrix} \rho_{\max} \mathbf{1}_{M_2} \\ -\varepsilon \mathbf{1}_{M_2} \end{bmatrix}, \quad (14)$$

where ε is a small positive number.

Next, define

$$\begin{aligned} \boldsymbol{\lambda} &= [1 \ \lambda \ \lambda^2 \ \lambda^3 \ \cdots \ \lambda^n]^T \\ \tilde{\boldsymbol{\lambda}}_1 &= [\mathbf{0}_{1 \times (m_1+m_2-1)(n+1)} \ \boldsymbol{\lambda}^T \ \mathbf{0}_{1 \times (m_1+m_2+2m_3)(n+1)}]^T, \\ \tilde{\boldsymbol{\lambda}}_2 &= [\mathbf{0}_{1 \times (M_1/2-1)(n+1)} \ \boldsymbol{\lambda}^T \ \mathbf{0}_{1 \times (M_1/2)(n+1)}]^T \end{aligned} \quad (15)$$

where λ is $\lambda(l)$, which denotes the radius of the virtual tube, the arguments are omitted for brevity. Then, define

$$\mathbf{A}_{2,j} = \mathbf{n}_b^T(l) \mathbf{t}_c(l) \mathbf{A}_{1,j} + \mathbf{n}_b^T(l) \mathbf{n}(l) \tilde{\boldsymbol{\lambda}}_1^T + \mathbf{n}_b^T(l) \mathbf{b}(l) \tilde{\boldsymbol{\lambda}}_2^T, \quad (16)$$

where $j = 1, 2, \dots, m_1$. Here, $\mathbf{A}_{2,j}$ is utilized to represent $v_{\text{obj}}(l)$ in Eq. (19) of the manuscript.

Subsequently, based on Eq. (24) of the manuscript and the second inequality in Eq. (26) of the manuscript, nonlinear inequality constraints of problem (32) in the manuscript are derived as

$$f_{k,j}(\mathbf{x}) \leq 0, k = 1, 2, 3, j = 1, 2, \dots, M_2. \quad (17)$$

Here

$$\begin{aligned}
f_{1,j}(\mathbf{x}) &= (\mathbf{B}_{3,j}\mathbf{x})^2 + (\mathbf{B}_{3,M_2+j}\mathbf{x})^2 + (\mathbf{B}_{3,2M_2+j}\mathbf{x})^2 - v_{\max}^2, \\
f_{2,j}(\mathbf{x}) &= \left((\mathbf{B}_{4,j}\mathbf{x})^2 + (\mathbf{B}_{4,M_2+j}\mathbf{x})^2 + (\mathbf{B}_{4,2M_2+j}\mathbf{x})^2 \right) - \left(\left(\frac{a_v}{\mathbf{B}_{3,j}\mathbf{x}} \right)^2 + \left(\frac{a_v}{\mathbf{B}_{3,M_2+j}\mathbf{x}} \right)^2 + \left(\frac{a_v}{\mathbf{B}_{3,2M_2+j}\mathbf{x}} \right)^2 \right), \\
f_{3,j}(\mathbf{x}) &= \left((\mathbf{B}_{5,3M_2+j}\mathbf{x})^2 + (\mathbf{B}_{5,4M_2+j}\mathbf{x})^2 + (\mathbf{B}_{5,5M_2+j}\mathbf{x})^2 \right) - \left(\left(\frac{a_\rho}{\mathbf{B}_{3,j}\mathbf{x}} \right)^2 + \left(\frac{a_\rho}{\mathbf{B}_{3,M_2+j}\mathbf{x}} \right)^2 + \left(\frac{a_\rho}{\mathbf{B}_{3,2M_2+j}\mathbf{x}} \right)^2 \right).
\end{aligned}$$

Next, according to *product rule for divergence*, Eq. (31) of the manuscript is derived as

$$\nabla_q \cdot (\rho \mathbf{v} S) = \rho S \nabla_q \cdot \mathbf{v} + \nabla_q (\rho S)^T \cdot \mathbf{v} = \rho S \nabla_q \cdot \mathbf{v} + \rho (\nabla_q S)^T \cdot \mathbf{v} + S (\nabla_q \rho)^T \cdot \mathbf{v} = 0, \quad (18)$$

where the arguments (l, d_n, d_b) are omitted for simplicity. Therefore, the nonlinear equality constraints of problem (32) in the manuscript are derived as

$$\begin{aligned}
h_j(\mathbf{x}) &= S(l) (\mathbf{B}_{1,j}\mathbf{x})^T (\mathbf{B}_{2,j}\mathbf{x}) + S'(l) (\mathbf{B}_{1,j}\mathbf{x})^T (\mathbf{B}_{7,j}\mathbf{x}) \\
&\quad + S(l) [(\mathbf{B}_{6,3M_2+j}\mathbf{x})^T (\mathbf{B}_{3,j}\mathbf{x}) + (\mathbf{B}_{6,4M_2+j}\mathbf{x})^T (\mathbf{B}_{3,M_2+j}\mathbf{x}) + (\mathbf{B}_{6,5M_2+j}\mathbf{x})^T (\mathbf{B}_{3,2M_2+j}\mathbf{x})] \\
&= 0,
\end{aligned} \quad (19)$$

where $j = 1, 2, \dots, M_2$.

Considering the relationship between the segments, the continuity conditions between adjacent segments are imposed as $\mathbf{v}_{d,j} = \mathbf{v}_{d,j-1}$, $\frac{\partial \mathbf{v}_{d,j}}{\partial l} = \frac{\partial \mathbf{v}_{d,j-1}}{\partial l}$, $\frac{\partial \mathbf{v}_{d,j}}{\partial d_n} = \frac{\partial \mathbf{v}_{d,j-1}}{\partial d_n}$, $\frac{\partial \mathbf{v}_{d,j}}{\partial d_b} = \frac{\partial \mathbf{v}_{d,j-1}}{\partial d_b}$, where $j = 2, 3, \dots, M_2$, which is equivalent to $v_{d,l,j} = v_{d,l,j-1}$, $\frac{\partial v_{d,l,j}}{\partial l} = \frac{\partial v_{d,l,j-1}}{\partial l}$, where $j = 2, 3, \dots, m_1$. The formulations for the d_n and d_b dimensions are analogous. For ρ , the above relationship exists similarly. They can be derived as

$$\begin{aligned}
\mathbf{C}_1 \mathbf{x} &= \mathbf{0}_{M_1}, \\
\mathbf{C}_2 \mathbf{x} &= \mathbf{0}_{M_1}.
\end{aligned} \quad (20)$$

Consequently, according to Eq. (20) of the manuscript and Eqs. (14)-(20) above, the optimization problem (32) of the manuscript can be discretized as

$$\begin{aligned}
&\min \sum_{i=1}^{m_1} \left(\frac{\omega_t}{\mathbf{A}_{1,i}\mathbf{x}} + \omega_\rho |\mathbf{P}_i| + \omega_v |\mathbf{A}_{2,i}\mathbf{x}| \right) \\
&\text{s.t. } \mathbf{B}\mathbf{x} \leq \mathbf{b} \\
&\quad \mathbf{C}\mathbf{x} = \mathbf{0} \\
&\quad f_{k,j}(\mathbf{x}) \leq 0 \\
&\quad h_j(\mathbf{x}) = 0.
\end{aligned} \quad (21)$$

where $k = 1, 2, 3$, $j = 1, 2, \dots, M_2$, $\mathbf{P}_i = \mathbf{B}_{1,i}\mathbf{x} - \rho_0 \left(\frac{S(l)}{S_0} \right)^{-\alpha}$, $\mathbf{B} = [\mathbf{B}_1; -\mathbf{B}_1]$, $\mathbf{b} = [\rho_{\max} \mathbf{1}_{M_2}; -\varepsilon \mathbf{1}_{M_2}]$, $\mathbf{C} = [\mathbf{C}_1, \mathbf{C}_2]^T$.

Supplementary Appendix C. The Proof of Proposition 2. This appendix provides a detailed proof of *Proposition 2* in Section III-C of the manuscript.

The proposition is proved by constructing a specific class of solutions where the vector field is steady, one-dimensional along the generating curve, and uniform across the cross-section. Let the velocity field be purely tangential to the generating curve with a constant magnitude $v_0 > 0$, that is, in the Frenet coordinate system.

$$\mathbf{v}_d(l, d_n, d_b) = [v_0, 0, 0]^T \quad (22)$$

where $\mathbf{t}_c(l)$ is the unit tangent vector of the generating curve. Since $\mathbf{v}_d(l, d_n, d_b)$ here depends solely on l , the notation $\mathbf{v}_d(l, d_n, d_b)$ is simplified to $\mathbf{v}_d(l)$ in the subsequent analysis. Let the density field be uniform across the cross-section,

$$\rho_d(l, d_n, d_b) = \rho_d(l). \quad (23)$$

Substituting Eqs. (22) and (23) into the continuity equation (31) of the manuscript yields $\frac{\partial}{\partial l}(\rho_d(l)v_0 S(l)) = 0$, that is $\rho_d(l)v_0 S(l) = C$, where C is a constant. Thus, density is determined by

$$\rho_d(l) = \frac{C}{v_0 S(l)}. \quad (24)$$

Other constraints can be satisfied by carefully selecting the constants v_0 and C . According to constraint (24) of the

manuscript, since $\|\mathbf{v}_d(l)\| = v_0$, the constraint holds if $v_0 \leq v_{max}$. Regarding constraint (26) of the manuscript, since $\mathbf{v}_d(l)$ is constant in magnitude, $\|\nabla_q \mathbf{v}_d(l)\|_F = 0$. The constraint $\|\nabla_q \mathbf{v}_d(l)\|_F \leq \|a_v/\mathbf{v}_d(l)\|$ becomes $0 \leq a_v/v_0$, which holds for any $a_v > 0, v_0 > 0$. Based on Eq. (24), the constraint $\rho_d(l) \leq \rho_{max}$ implies $\frac{C}{v_0 S(l)} \leq \rho_{max}$. Since $S(l) \geq S_{min}$, it suffices to choose C such that

$$C \leq v_0 S_{min} \rho_{max}. \quad (25)$$

The gradient of the density is dominated by the variation of the cross-section. If $\Delta_S = 0$, then $S'(l) = 0$, and the density-gradient constraint is automatically satisfied. If $\Delta_S > 0$, differentiating Eq. (24) with respect to l ,

$$\|\nabla_q \rho_d(l)\| = \left| \frac{\partial \rho_d}{\partial l} \right| = \left| -\frac{C S'(l)}{v_0 S(l)^2} \right|. \quad (26)$$

The constraint $\|\nabla_q \rho_d(l)\| \leq \|a_\rho/v_d(l)\|$ requires $\frac{C|S'(l)|}{v_0 S(l)^2} \leq \frac{a_\rho}{v_0}$, that is

$$C \leq \frac{a_\rho S(l)^2}{|S'(l)|}. \quad (27)$$

Since $S(l)$ is smooth and bounded below, the term $\frac{S(l)^2}{|S'(l)|}$ has a positive lower bound $\delta = \frac{S_{min}^2}{\Delta_S} > 0$. Thus, feasibility is guaranteed if

$$C \leq a_\rho \delta. \quad (28)$$

In summary, according to Eqs. (25) and (28), by selecting v_0 and C subject to

$$\begin{aligned} 0 < v_0 &\leq v_{max}, \\ 0 < C &\leq \min(v_0 S_{min} \rho_{max}, a_\rho \delta), \end{aligned} \quad (29)$$

all constraints in the optimization problem (32) of the manuscript are satisfied. This proves that the feasible set is nonempty.